Few-Shot SAR Target Recognition Through Meta-Adaptive Hyperparameters Learning for Fast Adaptation

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***Abstract*— In synthetic aperture radar automatic target recognition (SAR-ATR), the limitations of imaging environment and observation conditions make it challenging to acquire a substantial amount of high-value targets, resulting in a severe shortage of datasets. This scarcity leads to poor performance and instability in few-shot SAR target recognition. To address these shortcomings, this article proposes meta-adaptive stochastic gradient descent (Mada-SGD), a novel inner loop parameter update approach based on meta-adaptive hyperparameter learning. By considering the correlation information between multiple update steps, Mada-SGD learns the weight distribution information of initialization parameters across previous and current update steps, akin to a memory mechanism. This approach enhances feature extraction and representation ability for few-shot SAR targets. In addition, an adaptive hyperparameter update strategy is introduced to simultaneously learn the initialization, weight factor, update factor, and update direction in the meta-learner. This effectively resolves parameter updating issues in meta-learning models while improving fast adaptation for few-shot SAR targets. Experimental results on the specialized moving and stationary target acquisition and recognition few-shot learning (MSTAR-FSL) dataset demonstrate that Mada-SGD outperforms the latest few-shot SAR target recognition model in terms of SAR target recognition performance, validating its advancement and superiority.**

***Index Terms*— Automatic target recognition (ATR), deep learning, fast adaptation, few-shot learning (FSL), meta-learning, synthetic aperture radar (SAR).**

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I. Introduction

SYNTHETIC aperture radar (SAR) is an active remote sensing detection device that offers high-resolution imaging capabilities, enabling it to operate effectively in all-day and all-weather conditions. It can also penetrate cover and camouflage [1], [2], [3], making it highly suitable for automatic target recognition (ATR) tasks. SAR-ATR represents a critical branch in the field of radar signal interpretation and has not yet been fully resolved [4], [5]. One of the primary challenges lies in the scarcity of sufficient datasets due to varying imaging environments or confidentiality concerns, further resulting in an imbalance presence of high-quality SAR targets [6], [7], [8]. Therefore, research on SAR target recognition under the condition of limited samples stands as a key technology for enhancing the performance of current SAR-ATR systems.

Few-shot learning (FSL) aims to achieve robust generalization performance for new category targets using only a small number of labeled samples. It has been widely adopted in various fields, including computer vision and hyperspectral image processing [9], [10], [11], [12]. Currently, FSL can be categorized into three main directions: transfer learning (TL) [13], [14], data augmentation [15], [16], and meta-learning [17], [18]. The TL-based approaches involve pretraining on a large dataset to acquire prior knowledge, which is then fine-tuned with a small number of samples for effective knowledge transfer. Data augmentation methods leverage powerful generative models such as generative adversarial networks (GANs) to learn and generate new sample data. By combining newly generated samples with known ones, a larger dataset is created, enhancing the completeness of the target sample distribution, and facilitating subsequent training and classification. Meta-learning, also known as learning to learn, enables network models to quickly adapt to different tasks by constructing a series of task spaces. Even with limited samples, it exhibits strong target recognition performance and task generalization ability. Based on different learning mechanisms, current meta-learning FSL methods can be classified into three categories: optimization-based, model-based, and metric-based [19], [20]. The optimization-based methods transform the FSL problem into a two-level loop optimization problem. Through inner loop optimization, the

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meta-learning algorithm updates the learnable parameters in the meta-learner, such as initialization weights, learning rates, and update directions. Based on this, outer loop optimization is used to compute losses and learn the parameters of the base learner. A highly successful optimization-based approach is model-agnostic meta-learning (MAML) [21], which acquires a favorable initialization parameter and can rapidly adapt prior knowledge from the training set to the test target with just one iteration update. Motivated by this, subsequent researchers have proposed a series of enhanced algorithms focusing on updating step size, update strategy, etc. [22], [23], [24], [25], resulting in remarkable FSL performance improvements. The model-based methods primarily use recurrent neural networks (RNNs), such as long short-term memory (LSTM) and gated recurrent unit (GRU), to directly learn the internal relationship between samples in the training set and labels. They effectively preserve sample–label relationships using memory units and directly outputting a predictable network model. Algorithms in this category mainly include memory-augmented neural networks [26], meta-networks [27] meta-learner LSTM [28], etc. On the other hand, the metric-based methods concentrate on learning an efficient feature mapping space where different categories of targets can be well-represented while predicting targets through calculating similarity between training and testing samples. Typical examples encompass Siamese network [29], prototypical network [30], and matching network [31].

Regarding the aforementioned SAR target FSL (SAR-FSL) problems, although the dependency on target samples can be partially alleviated through the TL and data augmentation methods, they still possess certain limitations.

1) The TL-based methods still necessitate a sufficient number of target samples to fine-tune model parameters for ensuring recognition performance. Moreover, they require a high level of domain-specific knowledge, and their efficacy in tasks with significant domain differences cannot be guaranteed. The need for high-quality targets remains urgent and necessary, resulting in poor generalization performance.

2) Current data augmentation approaches ignore the imaging mechanism of SAR targets and merely learn from a limited number of target samples to generate an extensive set of new targets. This strategy theoretically resolves the issue of class imbalance caused by insufficient target samples through generating a new batch of target samples using a generative model; however, it lacks rigor and reasonableness from the perspective of SAR imaging mechanism. SAR target images adhere to strict physical scattering mechanisms and are sensitive to azimuth and depression angle. Consequently, under different imaging environments, the imaging results for the same target may vary significantly. In other words, the newly generated target samples produced by the data augmentation methods may not conform to the strict physical scattering mechanisms or exist in real-world situations. The unique learning mechanism of meta-learning makes it naturally suitable for addressing the challenge of SAR-FSL. And it has emerged as a prominent research area in recent years [32], [33], [34], [35]. In the field of metric-based SAR-FSL research, Gao et al. [36] proposed an approach that combined a multifeature fusion network (MFFN) and a weighted distance classifier (WDC). This method effectively mitigates the issues arising from large intraclass variance and high interclass similarity within the feature space of the traditional CNN-based models, thereby enhancing the representation ability for few-shot SAR targets. Wang et al. [37], on the other hand, developed an attribute-guided multiscale prototypical network (AG-MsPN) to comprehensively exploit subband decomposition features specific to SAR targets. Subband decomposition technique is used to extract rich amplitude and phase information from complex-valued SAR targets in both the azimuth and range directions. By leveraging this technique, more discriminative features can be learned for few-shot target recognition. In addition, to enhance the representation performance of SAR target features in the embedding space for subsequent classification, Wang et al. [38] investigated a hybrid inference network (HIN) that combines inductive and transductive inference to fully exploit both the Euclidean distance and label manifold structure for FSL and classification purposes. Yang et al. [39] built a mixed loss graph attention network (MGA-Net) that integrates data augmentation, attention mechanism, and mixed loss to alleviate the issue of data scarcity in SAR-FSL at both the data and algorithm levels. Their approach demonstrates high target recognition accuracy on the moving and stationary target acquisition and recognition (MSTAR) and OpenSARShip datasets, while exhibiting strong robustness against depression angle variations. Furthermore, Sun et al. [40] tackled the decline in recognition performance caused by the absence of target attitude angles in geospatial target SAR images by incorporating a scattering extraction model (SEM). They introduced a meta-learning architecture called the scattering characteristic analysis network (SCAN), which effectively enhances feature expression for SAR targets through comprehensively learning of angle-guided image domain features and scattering domain features.

To the best of our knowledge, however, most current SAR-FSL methods are metric-based meta-learning algorithms due to their comprehensibility and versatility. Few researchers focus on the optimization-based SAR-FSL methods, let alone the model-based approaches. Fu et al. [41] proposed an optimization-based meta-learning framework, called MSAR, to automatically learn a good initialization and update strategy for solving the SAR-FSL problem and achieving satisfactory results. In addition, Wang et al. [42] explored an SAR-FSL model that integrates probabilistic inference and meta-learning, effectively transferring simulated prior knowledge to real SAR targets and enhancing the ability to recognize few-shot targets. However, these aforementioned SAR-FSL methods have two deficiencies. First, in the inner loop update process, only the initialization parameters of the current update step are used as prior knowledge without fully considering the potential correlation information between multiple update steps. Second, only the initialization weights in the meta-learner are learned and optimized while ignoring the impact of learning rates, update directions, and other

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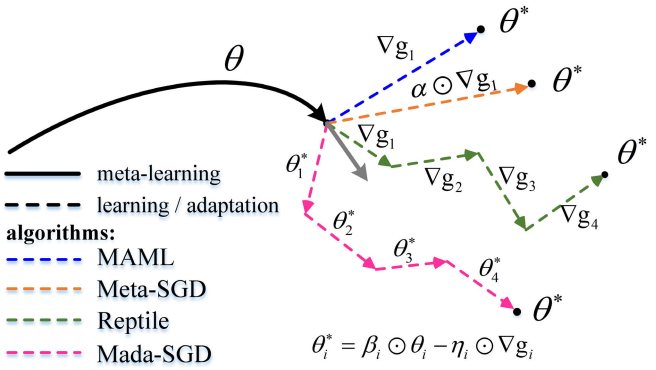


Fig. 1. Schematic of fast adaptation of Mada-SGD and classic MAML, Reptile, and Meta-SGD algorithms.

parameters on prior knowledge extraction. This leads to insufficient representation ability for few-shot SAR target features and poor generalization performance.

To address the above issues, this article proposes an SAR-FSL method that leverages a meta-adaptive hyperparameter update strategy to enhance the representation capability of few-shot SAR target features and achieve precise and stable few-shot target recognition. This update strategy is obtained through meta-learning and resembles stochastic gradient descent (SGD) in supervised learning, and thus, it is called meta-adaptive SGD (Mada-SGD). Compared with the conventional optimization-based meta-learning algorithms such as MAML [21], Reptile [23], and Meta-SGD [24], Mada-SGD learns not only a good initialization θ but also the weight factor, update factor, and update direction in the meta-learner. This effectively accelerates parameter updates and convergence speed while improving model's representation ability for SAR targets. Furthermore, by applying weight factors to multiple previous and current update steps, comprehensive correlation information between multiple inner loop update steps is considered to learn a better initialization for fast and stable parameters adaptation. Fig. 1 illustrates the fast adaptation of the proposed Mada-SGD. In summary, the main contributions of this article are as follows.

1) This article proposes a novel optimization-based meta-learning algorithm for SAR target FSL. Unlike most of the current SAR-FSL methods that use metric learning approaches, Mada-SGD is a completely new optimization-based SAR-FSL approach that effectively addresses the issues of limited samples and inadequate feature expression in SAR-ATR tasks caused by the challenges in obtaining high-quality targets.
    a) To achieve rapid generalization of prior knowledge to few-shot targets, this article presents an adaptive hyperparameter update strategy to automatically learn the weight factor, update factor, and update direction in the meta-learner, thereby effectively improving the fast adaptation ability of the meta-learner toward few-shot SAR targets.
    b) In addition, by imposing weight constraints on multiple update steps in the inner loop optimization process and comprehensively learning correlation information between these steps, a stable and efficient initialization parameter θ is obtained. This improves the representation ability of target features.

The remaining content of this article is organized as follows. Section II provides an introduction to the fundamental principles and concepts of meta-learning. Section III presents the derivation of the principle and formula for the proposed Mada-SGD method. Section IV encompasses experimental analysis and discussion, while Section V concludes with a summary and prospects for future research.

II. Preliminary

This section provides a concise overview and analysis of key concepts, principles, and current challenges in SAR-FSL, including the production of SAR target meta-dataset and the overall training and testing processes using Mada-SGD.

A. FSL in SAR-ATR

In SAR-ATR, the scarcity of adequate target sample data is a prevalent phenomenon. Hence, it is imperative and crucial to develop high-performance SAR target recognition methods in scenarios with limited samples. Traditional supervised learning necessitates a substantial number of diverse target training samples to acquire knowledge about the mapping from the input to the output: f, subsequently enabling identification and verification of targets in the testing set. In supervised learning, both the training set and the testing set share identical class labels for targets, that is, the target categories in the testing set align with those present in the training set [2], [4], [11]. SAR-FSL aims to address this issue by recognizing SAR targets when only a few samples are available. Due to an insufficient number of target samples, it becomes impractical to directly learn enough discriminative features through network training for accurate few-shot target recognition. Therefore, FSL initially learns discriminative features using a larger number of category targets and then uses only few-shot target samples (typically one-shot or five-shot) to generalize the model's discriminative ability toward new targets on the testing set [21], [29], [33]. Fig. 2 illustrates the distinction between traditional supervised learning and FSL. It can be observed that there is no overlap between the training set and the testing set in FSL and they are independent entities. In addition, it should be noted that within FSL's training set exists both support set and query set components which aid in facilitating learning regarding training targets.

B. Definition of Meta-Task

The technique of meta-learning is an FSL approach that uses learning tasks as the fundamental unit. Typically, a meta-dataset comprises the meta-training set and the meta-testing set, which consists of tasks sampled from a collection of target tasks. The tasks within the meta-training set are used for training the meta-learning model and extracting prior knowledge, while those in the meta-testing set are used to validate and evaluate the performance of the meta-learning model. In meta-learning, the tasks sampled for training are referred to as meta-training tasks, whereas new tasks used for

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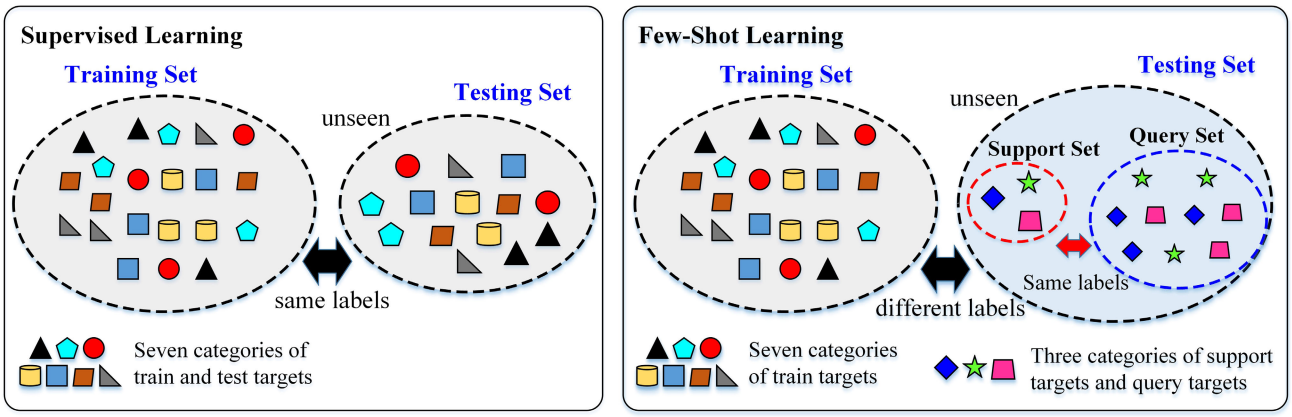


Fig. 2. Difference between traditional supervised learning and few-shot learning: (a) supervised learning and (b) few-shot learning.

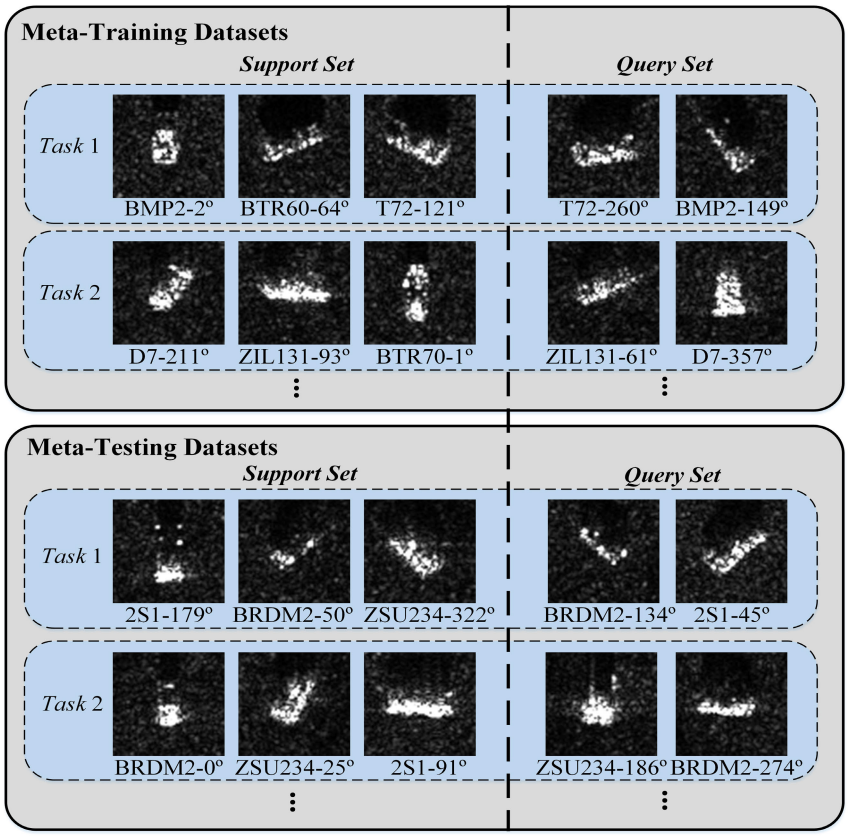


Fig. 3. Diagram of meta-dataset SAR-FSL three-way, one-shot task allocation.

testing are known as meta-testing tasks. Each task consists of a support set and a query set, resembling the training set and testing set in supervised learning, respectively. Furthermore, it can be described as an N-way, K-shot problem, where N represents the number of target categories and K denotes the number of targets within each category. Therefore, an N-way, K-shot task can be formulized as follows:

$$\text{Task} = \begin{cases} \text{Sample } N \text{ categories}, & N \leq C \\ \text{Sample } K \text{ target each category} \end{cases} \tag{1}$$

where C represents the total number of target categories in both the meta-training and meta-testing datasets. For an N-way, K-shot task, it comprises $N \times K + Q$ target samples, where Q denotes the number of targets in the query set. Fig. 3 displays an example of a meta-dataset composed of three-way, one-shot tasks from the MSTAR dataset. Each task in the support set consists of three categories of targets, with only one target sample per category, and the targets in the query set are derived from these target categories present in the support set, containing a total of two targets. In SAR-FSL, N and K are typically determined based on specific circumstances and task requirements. In this article, we adopt a combination of $N = 3$, $K = 1$ or 5 for training and testing purposes, further details regarding this configuration can be found in Section IV.

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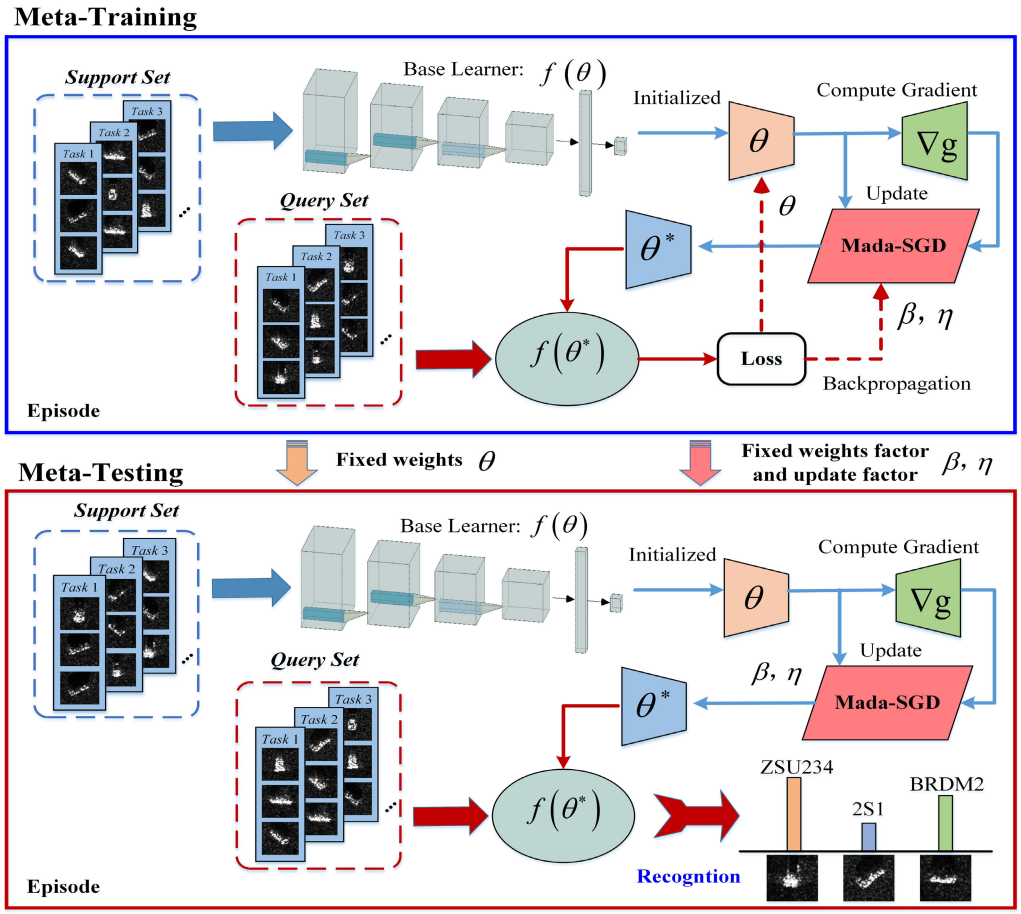


Fig. 4. Mada-SGD meta-training and meta-testing process.

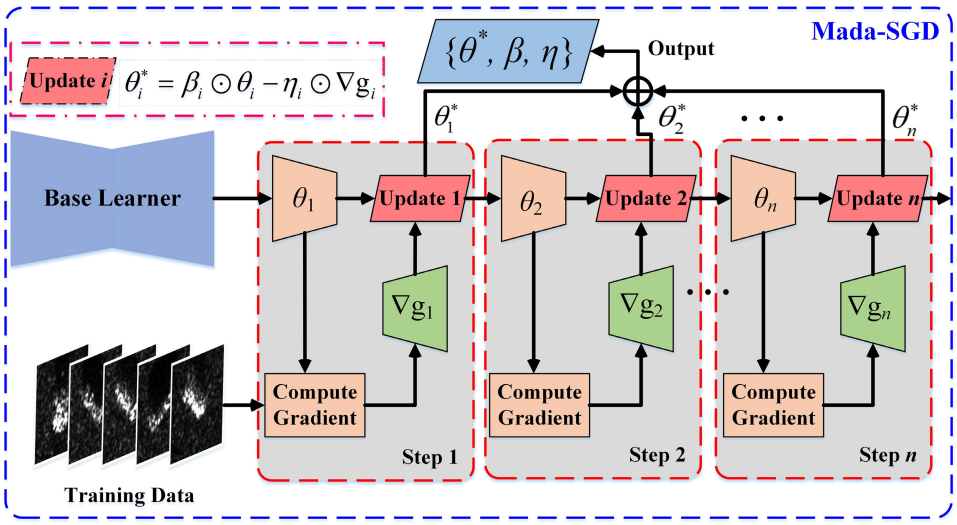


Fig. 5. Meta-adaptive hyperparameters' update strategy of Mada-SGD.

C. Meta-Training and Meta-Testing

The FSL, as an application of meta-learning in the field of supervised learning, primarily achieves robust generalization and fast adaptability across tasks by acquiring relevant objectives from a diverse range of tasks and guiding the learning process for new tasks. Meta-learning encompasses two stages: meta-training and meta-testing, which are used to address limited target sample recognition tasks. In addition, "episode" training is adopted to learn and update meta-parameters. Taking Mada-SGD as an example, Fig. 4 exhibits its execution process during both the meta-training and meta-testing stages. During the meta-training stage, the base learner focuses on learning generalizable knowledge and parameters across all the tasks (e.g., θ^*). The meta-learner generalizes training results from multiple base learners' tasks, learns parameter update strategies applicable to all the meta-tasks, and performs rapid and accurate inference on new

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tasks to update base learners and initialize parameters (e.g., initialization parameters θ, learning rates η, update rules). Furthermore, this two-level optimization process is repeated to complete the training of the meta-learner.

In the meta-testing stage, the initialization parameter θ, update factor η, and weight factor β are frozen and not updated by the meta-learner to facilitate fast adaptation for the targets in the meta-testing set and their subsequent recognition of new targets. First, the support set targets within the meta-testing set are used to learn and update the parameters specific to recognizing new targets. Subsequently, these fast adaptation parameters are input into the base learner for recognizing and evaluating the new targets within the query set. Throughout this testing process, fine-tuning of the base learner is achieved using N-way, K-shot tasks typically associated with using targets from the support set. The advantage of using "episode" training strategy lies in its ability to gather diverse learning tasks during training while simultaneously acquiring common features related to different combinations of tasks such as sample similarity comparison and extraction of universal characteristics. Consequently, when confronted with unseen tasks involving new targets, FSL models trained through this learning mechanism exhibit strong generalization capabilities.

III. Mada-SGD Algorithm

The commonly used optimization-based SAR target FSL approaches solely focus on optimizing the initialization parameters of the base learner, disregarding the influence of the update strategy on the meta-learning outcomes. In this section, we propose an SAR target FSL method based on a meta-adaptive hyperparameter update strategy, aiming to enhance the expressive capability of the target while simultaneously learning the initialization weights. The training and testing processes of our proposed Mada-SGD are shown in Fig. 4. Subsequently, we will provide a detailed introduction to the principle and implementation of Mada-SGD.

A. Meta-Learner

Before describing the parameter update process of the proposed Mada-SGD, we first introduce a general optimization problem setting in meta-learning. We assume that the task space generated by all the targets in meta-learning is $p(\mathcal{T})$, and each learning task can be sampled from this space to obtain $\mathcal{T}_i \sim p(\mathcal{T})$. For an N-way, K-shot task $\mathcal{T}_i$, we first randomly sample K targets from $\mathcal{T}_i$ to form the support set $\mathcal{D}_i$, which is used for fast adaptation of the task. Then, we randomly sample the remaining samples of $\mathcal{T}_i$ to obtain the query set $\mathcal{D}_i^*$ for evaluating the model's generalization performance on new targets. The corresponding loss function for this evaluation is $\mathcal{L}_{\mathcal{T}_i}$. Finally, the loss function $\mathcal{L}_{\mathcal{T}_i}$ is used to adjust the model parameters to enhance its generalization ability across different tasks.

The goal of the meta-learner in the classic MAML and its variants [21], [23], [24] is to incorporate prior knowledge from the task into the initialization weights θ of the base learner $f(\theta)$, aiming to obtain an effective initialization position for fast adaptation to new tasks. Given a randomly sampled task $\mathcal{T}_i$, with its corresponding support set $\mathcal{D}_i$, loss function $\mathcal{L}_{\mathcal{T}_i}^{\mathcal{D}_i}$, and initialization parameters θ_i, the inner loop update of the meta-learner is

$$\boldsymbol{\theta}_i^* = \boldsymbol{\theta}_i - \alpha \cdot \nabla_\theta \mathcal{L}_{\mathcal{T}_i}^{\mathcal{D}_i}\left(f(\boldsymbol{\theta}_i), \boldsymbol{y}_i\right) \tag{2}$$

where α is the learning rate, y_i is the label of the targets in the support set $\mathcal{D}_i$, $\nabla_\theta \mathcal{L}_{\mathcal{T}_i}^{\mathcal{D}_i}(f(\theta_i))$ represents the gradient of the current task $\mathcal{T}_i$, and θ_i^* indicates the network weights after adapting to task $\mathcal{T}_i$. Next, to assess the generalization performance of the base learner under adapted network weights θ_i^*, we evaluate its performance on the query set $\mathcal{D}_i^*$. Typically, during testing, we update the initialization weights θ of the base learner by incorporating the prior knowledge from multiple different learning tasks to enhance cross-task learning generalization ability. Hence, this optimization process is also referred to as outer loop update with its parameter update formula represented as follows:

$$\boldsymbol{\theta} = \boldsymbol{\theta} - \lambda \cdot \nabla_{\boldsymbol{\theta}} \sum_{\mathcal{T}_i} \mathcal{L}_{\mathcal{T}_i}^{\mathcal{D}_i^*}\left(f\left(\boldsymbol{\theta}_i^*\right), \boldsymbol{y}_i^*\right) \tag{3}$$

where λ and y_i^* represent the parameter update rate and the target labels in the query set $\mathcal{D}_i^*$, respectively. Through the outer loop update optimization in the meta-learner, the prior knowledge adapted to specific tasks in the base learner is generalized to new tasks. During testing, only a simple one-step or multistep inner loop update on the new task is required to achieve learning and generalization for that task.

Typically, an optimizer comprises three components: initialization, update direction, and learning rate. Initialization is commonly set randomly to configure parameters, while the update direction heavily relies on gradient calculation. As for the learning rate, it is usually determined through manual settings. However, the above update rules may not be reliable and effective for FSL tasks that only involve a limited target samples compared with supervised learning tasks with abundant labeled samples for learning. This limitation arises from the fact that ordinary learning tasks can exploit adequate labeled samples to optimize and fine-tune an excellent optimizer, whereas in FSL tasks, there might not have enough samples to support this process adequately. Consequently, this article proposes a parameter adaptive update method based on meta-learning called Mada-SGD to automatically learn and configure the optimizer. The aim is to improve the feature representation ability for few-shot SAR targets. The specific calculation process is described as follows.

Assuming that the initialization parameter of the current update step is θ_i, then the weighted inner loop update is

$$\begin{aligned}\boldsymbol{\theta}_i^* &= \boldsymbol{\beta}_i \odot \left[\boldsymbol{\theta}_i - \boldsymbol{\alpha}_i \odot \nabla_{\boldsymbol{\theta}} \mathcal{L}_{\mathcal{T}_i}^{\mathcal{D}_i}\left(f(\boldsymbol{\theta}_i), \boldsymbol{y}_i\right)\right] \\ &= \boldsymbol{\beta}_i \odot \boldsymbol{\theta}_i - \boldsymbol{\beta}_i \odot \boldsymbol{\alpha}_i \odot \nabla_{\boldsymbol{\theta}} \mathcal{L}_{\mathcal{T}_i}^{\mathcal{D}_i}\left(f(\boldsymbol{\theta}_i), \boldsymbol{y}_i\right)\end{aligned} \tag{4}$$

where β_i and $\odot$ are the weight coefficient and Hadamard (elementwise) product, respectively. Let $\eta_i = \beta_i \odot \alpha_i$ in the above formula, we can obtain

$$\boldsymbol{\theta}_i^* = \boldsymbol{\beta}_i \odot \boldsymbol{\theta}_i - \boldsymbol{\eta}_i \odot \nabla_{\boldsymbol{\theta}} \mathcal{L}_{\mathcal{T}_i}^{\mathcal{D}_i}\left(f(\boldsymbol{\theta}_i), \boldsymbol{y}_i\right). \tag{5}$$

Then, considering the impact of multiple update steps on the initialization parameters, we introduce weight coefficients for

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multiple update steps to comprehensively learn the correlation information between them, allowing past prior knowledge to contribute to the current weight parameters. This improves the robust expression of target features. The specific operation is described as follows:

$$\boldsymbol{\theta}^* = \sum_i^N \boldsymbol{\theta}_i^* = \sum_i^N \left(\boldsymbol{\beta}_i \odot \boldsymbol{\theta}_i - \boldsymbol{\eta}_i \odot \nabla_{\boldsymbol{\theta}} \mathcal{L}_{\mathcal{T}_i}^{\mathcal{D}_i}\left(f(\boldsymbol{\theta}_i), \boldsymbol{y}_i\right)\right) \quad (6)$$

where β and η are the learnable parameters, representing the weight factor and update factor, respectively. The former controls the decay rate of the current weight parameters, while the latter controls the step size for current parameter updates. Mada-SGD proposes a novel adaptive parameter update strategy in inner loop update to simultaneously solve initialization, update direction, and learning rate in meta-learning optimizers. This strategy not only learns good initialization parameter but also learns weight factors and update factors between different inner loop update steps. It has the advantage of adaptively and comprehensively using correlation information between multiple inner loop update steps to improve SAR target feature extraction by the meta-learner and learn more robust prior knowledge. Fig. 5 illustrates the execution process of Mada-SGD. Combining (3), the outer loop parameter update formula is derived as follows:

$$(\boldsymbol{\theta}, \boldsymbol{\beta}, \boldsymbol{\eta}) = (\boldsymbol{\theta}, \boldsymbol{\beta}, \boldsymbol{\eta}) - \lambda \cdot \nabla_{(\boldsymbol{\theta}, \boldsymbol{\beta}, \boldsymbol{\eta})} \sum_{\mathcal{T}_i} \mathcal{L}_{\mathcal{T}_i}^{\mathcal{D}_i^*}\left(f\left(\boldsymbol{\theta}_i^*\right), \boldsymbol{y}_i^*\right). \quad (7)$$

B. Fast Adaptation From Regularization Perspective

The introduction of weight factor not only applies weight coefficients to the initialization parameters of multiple different update steps but also imposes specific regularization constraints on the update rule itself. This is because the weight factor scales the weights during fast adaptation, similar to a weight decay operation. This section derives the inner loop update rule of Mada-SGD from the perspective of regularization constraints, as depicted below.

Initially, the ℓ_2 regularization term, $(\kappa/2)\|\boldsymbol{\theta}_i\|$, is incorporated into the loss function $\mathcal{L}_{\mathcal{T}_i}^{\mathcal{D}_i}$. Then, the inner loop update formula can be obtained according to (2)

$$\begin{aligned}\boldsymbol{\theta}_i^* &= \boldsymbol{\theta}_i - \alpha \cdot \nabla_{\boldsymbol{\theta}}\left(\mathcal{L}_{\mathcal{T}_i}^{\mathcal{D}_i}\left(f(\boldsymbol{\theta}_i), \boldsymbol{y}_i\right) + \frac{\kappa}{2}\|\boldsymbol{\theta}_i\|\right) \\ &= \boldsymbol{\theta}_i - \alpha \cdot \left(\nabla_{\boldsymbol{\theta}} \mathcal{L}_{\mathcal{T}_i}^{\mathcal{D}_i}\left(f(\boldsymbol{\theta}_i), \boldsymbol{y}_i\right) + \kappa \cdot \boldsymbol{\theta}_i\right) \\ &= (1 - \alpha \cdot \kappa) \cdot \boldsymbol{\theta}_i - \alpha \cdot \nabla_{\boldsymbol{\theta}} \mathcal{L}_{\mathcal{T}_i}^{\mathcal{D}_i}\left(f(\boldsymbol{\theta}_i), \boldsymbol{y}_i\right) \end{aligned} \quad (8)$$

where κ represents the regularization coefficient. Let $\beta = (1 - \alpha \cdot \kappa)$, the above formula is equivalent to

$$\boldsymbol{\theta}_i^* = \beta \cdot \boldsymbol{\theta}_i - \alpha \cdot \nabla_{\boldsymbol{\theta}} \mathcal{L}_{\mathcal{T}_i}^{\mathcal{D}_i}\left(f(\boldsymbol{\theta}_i), \boldsymbol{y}_i\right). \quad (9)$$

Taking into account multiple update steps and calculating the weight and learning rate leads to the following inner loop update rule:

$$\boldsymbol{\theta}^* = \sum_i^N \left(\boldsymbol{\beta}_i \odot \boldsymbol{\theta}_i - \boldsymbol{\alpha}_i \odot \nabla_{\boldsymbol{\theta}} \mathcal{L}_{\mathcal{T}_i}^{\mathcal{D}_i}\left(f(\boldsymbol{\theta}_i), \boldsymbol{y}_i\right)\right). \quad (10)$$

Let $\boldsymbol{\eta}_i = \boldsymbol{\alpha}_i$, then the effects of (6) and (10) are completely equivalent. Both $\boldsymbol{\beta}_i$ and $\boldsymbol{\alpha}_i$ are learnable parameters that are automatically updated and solved by the meta-learner based on the learning tasks. Hence, to certain extent, Mada-SGD can be considered as a regularized meta-learning optimizer that comprehensively incorporates multistep update parameters. It not only considers the correlation between multiple update steps but also imposes a regularization constraint to each step to prevent model overfitting, thereby effectively generalizing prior knowledge to unseen targets.

C. Multilevel Inner Loop Update Strategy

To adapt to the FSL task of SAR targets in different scenarios, this section introduces a multi-level inner loop update strategy, namely, modelwise, layerwise, and elementwise, to effectively consider the fast adaptation ability under different computational complexities, as shown in Fig. 6. First, the inner loop update rule based on modelwise updates the parameters by applying a weight factor and an update factor to the initialization $\boldsymbol{\theta}$ and gradients $\nabla \boldsymbol{g}$, respectively. Under this update rule, there are S inner loop update steps with a total of learnable parameters for the weight factor β and update factor η being $2 \times S$. Second, the layerwise update rule applies a weight factor $\boldsymbol{\beta}$ and update factor $\boldsymbol{\eta}$ to each layer of the initialization $\boldsymbol{\theta}$ and gradients $\nabla \boldsymbol{g}$, respectively. The number of learnable parameters is $2 \times N \times S$, where N denotes the number of layers. Finally, the elementwise update rule independently multiplies each element of the initialization $\boldsymbol{\theta}$ and gradients $\nabla \boldsymbol{g}$ by a weight factor $\boldsymbol{\beta}$ and update factor $\boldsymbol{\eta}$, respectively. Its parameter amount is $2 \times M \times S$. It should be noted that M represents the number of parameters in the initialization weights. Elementwise Mada-SGD has superior learning ability due to its large number of learnable parameters; however, it also possesses higher computational complexity which may lead to overfitting when dealing with few-shot target data. Therefore, in the actual application process, it is imperative to carefully select the appropriate level of model parameter complexity based on task requirements to ensure optimal performance of Mada-SGD. Specifically, the selection of models for larger data volumes tends to favor those with more parameters, such as layerwise and elementwise models. Conversely, smaller data volumes are better suited for models with fewer parameters, such as modelwise models. Moreover, when considering task requirements, more complex tasks benefit from models with stronger learning capabilities (e.g., layerwise and modelwise), while simpler tasks can mitigate overfitting issues using weaker learning capabilities (e.g., modelwise). Algorithm 1 outlines the overall learning process of Mada-SGD.

D. Base Learner Architecture and Loss Function

In this article, ConvNet4 [21], [23], [24], [41] is adopted as the backbone network structure of the base learner to comprehensively evaluate the SAR target recognition performance of Mada-SGD under various test conditions. The network architecture is shown in Fig. 7. It can be observed that ConvNet4 primarily comprises an input

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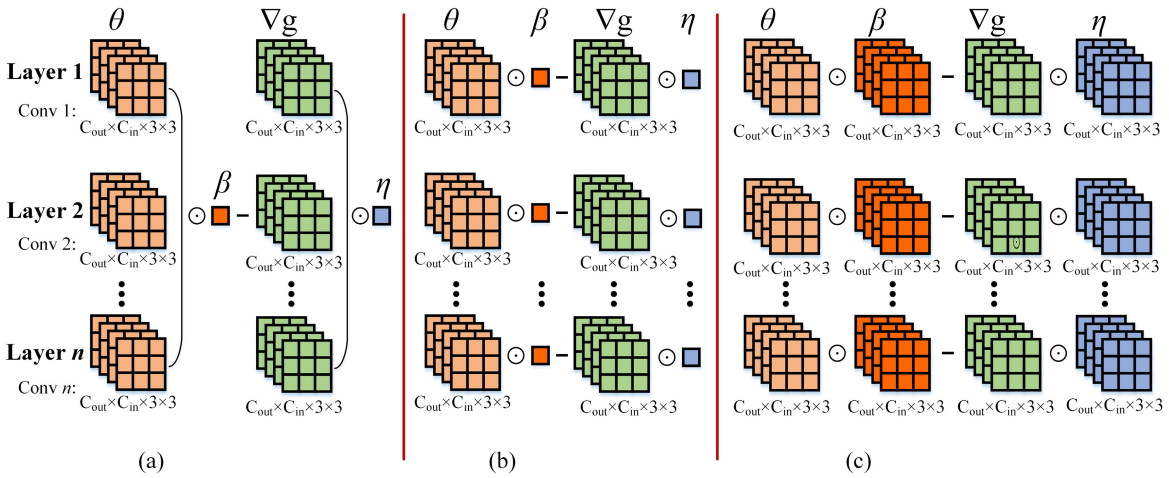


Fig. 6. Multilevel inner loop update rules for fast adaptation. (a) Modelwise. (b) Layerwise. (c) Elementwise.

Algorithm 1 Mada-SGD for SAR-FSL

Require: task distribution $p(\mathcal{T})$, learning rate λ.

1. **Initialize** $\boldsymbol{\theta}$, $\boldsymbol{\beta}$, $\boldsymbol{\eta}$ and S.
2. **while** not converged **do**
3. Sampling a batch of tasks $\mathcal{T}_i \sim p(\mathcal{T})$.
4. **for** each task $\mathcal{T}_i$ **do**
5. Sampling *support set* $\mathcal{D}_i$ and *query set* $\mathcal{D}_i^*$ from $\mathcal{T}_i$.
6. **for** inner loop step $j = 1$ to S **do**
7. Calculate train loss:

$$\mathcal{L}_{\mathcal{T}_i}^{\mathcal{D}_i}(f(\boldsymbol{\theta}), \boldsymbol{y}) \leftarrow \frac{1}{|\mathcal{D}_i|} \sum_{(x,y)\in\mathcal{D}_i} \mathcal{L}(f_{\boldsymbol{\theta}}(x), y)$$

8. Calculate task-adapted weights:

$$\boldsymbol{\theta}^* \leftarrow \sum_{j}^{N} (\boldsymbol{\beta} \odot \boldsymbol{\theta} - \boldsymbol{\eta} \odot \nabla_{\theta} \mathcal{L}_{\mathcal{T}_i}^{\mathcal{D}_i}(f(\boldsymbol{\theta}), \boldsymbol{y}))$$

9. **end for**
10. Calculate test loss:

$$\mathcal{L}_{\mathcal{T}_i}^{\mathcal{D}_i^*}(f(\boldsymbol{\theta}^*), \boldsymbol{y}^*) \leftarrow \frac{1}{|\mathcal{D}_i^*|} \sum_{(x,y)\in\mathcal{D}_i^*} \mathcal{L}(f_{\boldsymbol{\theta}^*}(x), y)$$

11. Update: $(\boldsymbol{\theta}, \boldsymbol{\beta}, \boldsymbol{\eta}) \leftarrow (\boldsymbol{\theta}, \boldsymbol{\beta}, \boldsymbol{\eta}) - \lambda \cdot \nabla_{(\boldsymbol{\theta},\boldsymbol{\beta},\boldsymbol{\eta})} \sum_{\mathcal{T}_i} \mathcal{L}_{\mathcal{T}_i}^{\mathcal{D}_i^*}(f(\boldsymbol{\theta}^*), \boldsymbol{y}^*)$
12. **end for**
13. **end while**
14. **output:** $\{\boldsymbol{\theta}, \boldsymbol{\beta}, \boldsymbol{\eta}\}$

layer, four convolutional layers, and a classifier. Within the convolutional layers, targets undergo sequential processing involving: convolution → batch normalization → ReLU activation → pooling operations, to progressively abstract and extract features. For handling SAR target FSL tasks with diverse complexities, a 3 × 3 convolution kernel with 64 channels is used by the base learner.

Mada-SGD can adaptively learn the initialization and hyperparameter update rules in the meta-learner from the training tasks, enabling the application of learned prior

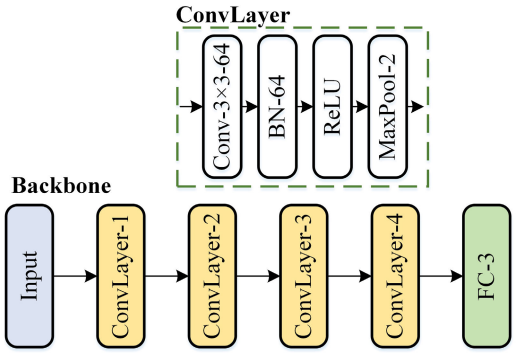


Fig. 7. Network architecture of base learner-ConvNet4.

knowledge to tasks in the testing set. The initialization parameters serve as an effective starting position for learning with limited samples in the testing set, while the update rules ensure fast adaptation to the test targets through a few update steps, thereby significantly improving SAR-FSL's learning and representation capabilities. To explicitly train Mada-SGD, we first conduct fast adaptation on the support set of meta-training tasks train($\mathcal{T}$) using a base learner and then calculate the generalization error on the query set of meta-training tasks test($\mathcal{T}$) to simultaneously learn both initialization and update rules. Therefore, the objective function of Mada-SGD is to minimize the generalization error on the meta-testing tasks test($\mathcal{T}$), which can be expressed as:

$$\begin{aligned}\{\boldsymbol{\theta}, \boldsymbol{\beta}, \boldsymbol{\eta}\} &\leftarrow \min_{\boldsymbol{\theta},\boldsymbol{\beta},\boldsymbol{\eta}} E_{\mathcal{T}\sim p(\mathcal{T})}\left[\mathcal{L}_{\text{test}(\mathcal{T})}\left(\boldsymbol{\theta}^*\right)\right] \\ &= \min_{\boldsymbol{\theta},\boldsymbol{\beta},\boldsymbol{\eta}} E_{\mathcal{T}\sim p(\mathcal{T})}\left[\mathcal{L}_{\text{test}(\mathcal{T})}\left(\sum_{i}^{N}\left(\boldsymbol{\beta} \odot \boldsymbol{\theta} - \boldsymbol{\eta} \odot \nabla_{\boldsymbol{\theta}} \mathcal{L}_{\text{train}(\mathcal{T})} \times \left(f(\boldsymbol{\theta}), \boldsymbol{y}\right)\right)\right)\right]. \end{aligned} \tag{11}$$

In Mada-SGD, various forms of loss functions $\mathcal{L}(\cdot)$ can be selected for differentiated learning according to task requirements. In this article, we primarily adopt cross-entropy loss (CEL) to quantify the discrepancy between different tasks.

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TABLE I
CONFIGURATION INFORMATION OF THE MSTAR-FSL DATASET

	Target Categories	Serial No.	Target Number	Depression
Training Set	BMP2	9563	233	17º
	BTR60	k10yt7532	256	
	BTR70	c71	233	
	D7	92v13015	299	
	T62	a51	299	
	T72	132	232	
	ZIL131	e12	299	
Testing Set	2S1	b01	274, 299, 288, 303	15º, 17º, 30º, 45º, respectively
	BRDM2	e71	274, 298, 287, 303	
	ZSU234	d08	274, 299, 288, 303	

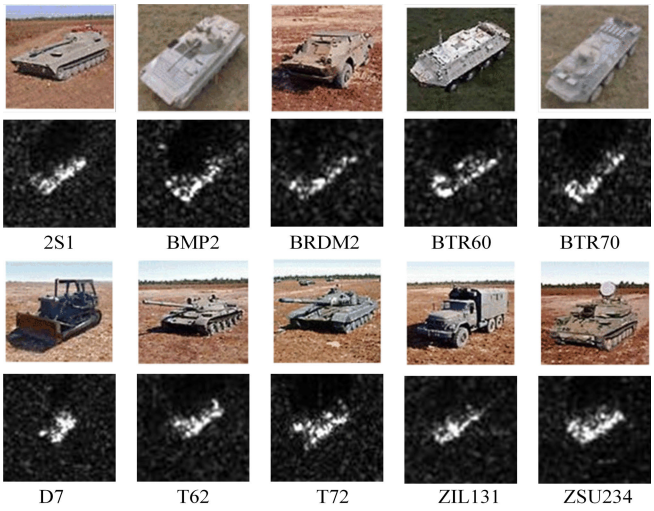


Fig. 8. Ten categories of targets in the MSTAR dataset. (Top) Optical images versus (bottom) SAR images.

The formula for CEL is presented as follows:

$$\mathcal{L}(\boldsymbol{y}, \boldsymbol{f}) = -\sum_{i=1}^{n} y_i \cdot \log(f_i), \quad i = 1, 2, \ldots, n \tag{12}$$

where $\boldsymbol{f}$ and $\boldsymbol{y}$ represent the predict label and true label, respectively. n indicates the number of SAR targets in meta-tasks.

IV. EXPERIMENTAL RESULTS' ANALYSIS AND DISCUSSION

This section mainly conducts the experimental testing analysis and discussion of our proposed Mada-SGD, encompassing three aspects: few-shot dataset preparation, performance evaluation, and model analysis.

A. Production and Preparation of MSTAR-FSL Dataset

The MSTAR dataset [3], [6], [39], [43], [44] is a benchmark dataset for SAR target evaluation supported by the U.S. Defense Advanced Research Projects Agency (DARPA). It adopts HH polarization, X-band frequency, with a resolution of 0.3 m × 0.3 m and a 1° spacing in 360° articulation imaging. The dataset contains ten categories of targets, and their corresponding optical and SAR images are shown in Fig. 8. To facilitate experimental testing and analysis based on literature [3], [6], [38], [39], [45] and specific requirements, this section produces a specialized MSTAR few-shot target recognition dataset called the MSTAR-FSL dataset. The configuration information is listed in Tables I and II. This article primarily uses the 3-way, K-shot tasks for both training and testing purposes. In this setup the meta-training set consists of samples from seven target categories, while the meta-testing set includes samples from three different target categories (2S1, BRDM2, and ZSU234). This division offers the following benefits.

1) By enlarging the size of the meta-training task space as much as possible due to only having ten target categories in MSTAR dataset, overfitting can be prevented while enhancing adaptability of meta-testing models across different tasks. The utilization of a three-way, K-shot task expands the space of meta-training tasks to $C_7^3 = 35$, surpassing other partitioning strategies; thus facilitating learning universal prior knowledge related to SAR targets.
2) Selecting specific targets such as 2S1, BRDM2, and ZSU234 for evaluation during meta-testing enables us

TABLE II
CONFIGURATION INFORMATION OF THE N-WAY, K-SHOT META-TASK FOR SAR-FSL

Task	Meta-Training		Meta-Testing	
	Support Set	*Query Set*	*Support Set*	*Query Set*
3-way, 1-shot	3*1=3	3*15=45	3*1=3	3*15=45
3-way, 5-shot	3*5=15	3*15=45	3*5=15	3*15=45

TABLE III
EXPERIMENT SETTINGS OF THE MSTAR-FSL DATASET

Experiment Types	Meta-Training Task		Meta-Testing Task	
	Task Num	Depression	Task Num	Depression
17ºD/15ºD	4000	17º	300	15º
17ºD/17ºD		17º		17º
17ºD/30ºD		17º		30º
17ºD/45ºD		17º		45º

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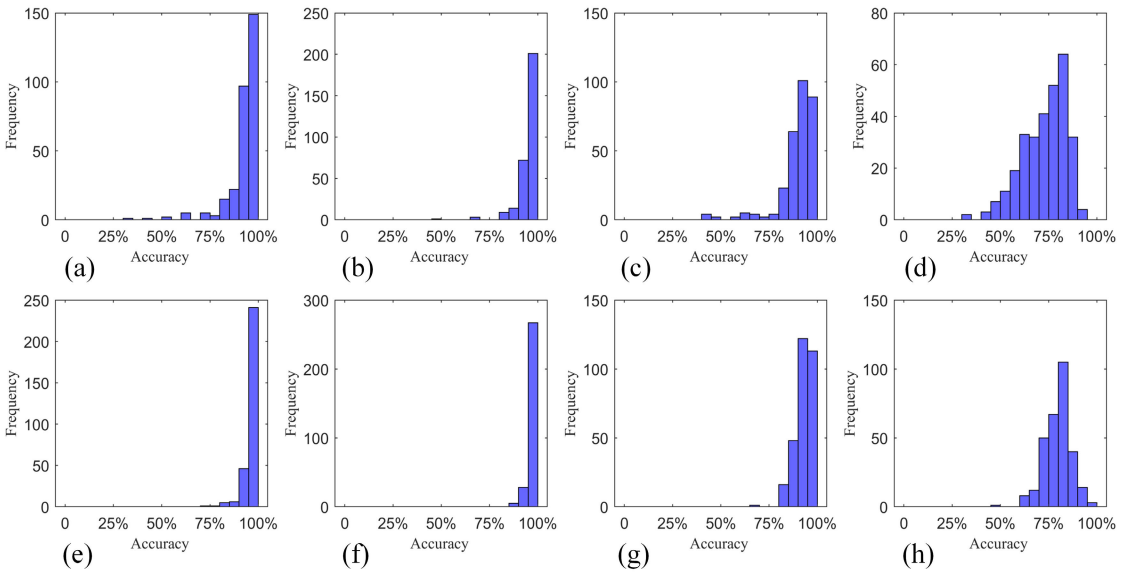


Fig. 9. Statistical histograms of the test accuracy for one-shot and five-shot under various depression angle configuration on the MSTAR-FSL dataset. (a) 17°D/15°D (one-shot). (b) 17°D/17°D (one-shot). (c) 17°D/30°D (one-shot). (d) 17°D/45°D (one-shot). (e) 17°D/15°D (five-shot). (f) 17°D/17°D (five-shot). (g) 17°D/30°D (five-shot). (h) 17°D/45°D (five-shot).

TABLE IV
QUANTITATIVE TEST RESULTS OF MADA-SGD TARGET RECOGNITION. MEAN ACCURACY IS SHOWN IN BOLD

Model	Base Learner	Experiment Type		Acc (%)			Std
		Dataset	Task	*Min*	*Max*	*Mean*	
Mada-SGD	ConvNet4	17°D/15°D	1-shot	31.11	100	**92.52**	0.0869
			5-shot	71.11	100	**96.19**	0.0373
		17°D/17°D	1-shot	48.89	100	**95.08**	0.0550
			5-shot	86.67	100	**97.27**	0.0256
		17°D/30°D	1-shot	40.00	100	**89.54**	0.0971
			5-shot	68.89	100	**92.74**	0.0424
		17°D/45°D	1-shot	31.11	93.33	**72.59**	0.1044
			5-shot	51.11	97.78	**79.39**	0.0692

to assess the impact of depression angle under extended operation conditions (EOCs). For a three-way, K-shot task, there are K samples ($K = 1$ or 5) for each target category in the support set, resulting in a total of $3 \times K$ SAR target samples while 15 samples in the query set, a total of $3 \times 15 = 45$ target samples.

B. Performance Evaluation

The experimental setup and configuration details are listed as follows: Windows 10 system, Python version 3.7, PyTorch version 1.9, CUDA version 10.2, Intel Core i7-7700K CPU, and Nvidia GeForce RTX 2060 (12 GB) GPU. The performance evaluation of the experiment mainly adopts three metrics: accuracy (Acc), standard deviation (Std), and loss error (Loss). The definitions of these metrics are as follows:

$$\text{Acc} = \frac{\text{Number of correct targets}}{\text{Number of total targets}} \tag{13}$$

$$\text{Std} = \sqrt{\frac{\sum_{\mathcal{T}_i}^{\mathcal{T}_n} \left(\text{Acc}_{\mathcal{T}_i} - \mu\right)^2}{n}} \tag{14}$$

where $\mathcal{T}_i$ represents the ith meta-testing task, $\text{Acc}_{\mathcal{T}_i}$ indicates the recognition accuracy of that task, and μ denotes the average recognition accuracy of n recognition tasks. The loss error is calculated using CEL as defined in (12). For meta-training set, we randomly sampled 4000 meta-tasks while for meta-testing we randomly sampled 300 meta-tasks to evaluate few-shot target recognition performance ensuring equal selection probability for each target. Furthermore, SAR-FSL experiments were conducted under different depression angle conditions as shown in Table III. For instance, 17°D/15°D represents that the depression angle of the meta-training target is 17° while it is 15° for the meta-testing target.

1) Recognition Results of Mada-SGD: This section primarily evaluates the performance of Mada-SGD and comprehensively examines the impact of different depression angle experimental configurations on the SAR-FSL recognition results. Through quantification and analysis of 300 meta-testing tasks, the experimental results are shown in Table IV. It can be observed that Mada-SGD has achieved outstanding SAR-FSL recognition results across four experimental tests. Specifically, in the one-shot testing task, the target recognition accuracy rates are as follows: 92.52%, 95.08%, 89.54%, and 72.59%. Furthermore, in the five-shot testing task, as the number of available target samples for learning

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TABLE V
AVERAGE CONFUSION MATRIX OF MADA-SGD

Experiment Type	True \ Predict	*3-way, 1-shot*				*3-way, 5-shot*			
		2S1	BRDM2	ZSU234	Recall (%)	2S1	BRDM2	ZSU234	Recall (%)
17°D/15°D	2S1	248.06	14.55	11.39	90.53	260.69	4.01	4.3	96.91
	BRDM2	10.41	256.1	7.49	93.47	5.68	259.14	4.18	96.33
	ZSU234	7.25	10.41	256.34	93.55	6.51	6.1	256.39	95.31
	Precision (%)	93.35	91.12	93.14	92.52	95.53	96.25	96.80	96.19
17°D/17°D	2S1	278.6	10.59	8.81	93.49	289.66	4.7	3.64	97.20
	BRDM2	5.43	287.54	5.03	96.49	4.17	289.92	3.91	97.29
	ZSU234	5.83	8.28	283.89	95.27	3.24	4.71	290.05	97.33
	Precision (%)	96.12	93.84	95.35	95.08	97.51	96.86	97.46	97.27
17°D/30°D	2S1	252.88	17.6	16.52	88.11	265.38	11.22	10.4	92.47
	BRDM2	12.25	262	12.75	91.29	11.1	265.95	9.95	92.67
	ZSU234	16.45	14.48	256.07	89.22	10.33	9.5	267.17	93.09
	Precision (%)	89.81	89.09	89.74	89.54	92.53	92.77	92.92	92.74
17°D/45°D	2S1	217.42	45.11	40.47	71.76	255.26	23.43	24.31	84.24
	BRDM2	40.00	222.2	40.8	73.33	49.09	220.65	33.26	72.82
	ZSU234	41.48	41.34	220.18	72.67	30.77	26.53	245.7	81.09
	Precision (%)	72.74	71.99	73.04	72.59	76.17	81.54	81.02	79.39

increases, Mada-SGD demonstrates improved recognition performance with target recognition accuracy rates reaching: 96.19%, 97.27%, 92.74%, and 79.39%, respectively. Table V displays the average confusion matrix of Mada-SGD under various experimental testing tasks. It is evidence that Mada-SGD exhibits satisfactory recognition performance for three different types of SAR targets (2S1, BRDM2, and ZSU234), without focusing solely on recognizing a specific type target. This leads to significant difference in recognition performance among different targets. In addition, the precision and recall rate analysis between different types of targets within the confusion matrix reveals that our proposed Mada-SGD maintains high recall rate and precision simultaneously, resulting in more stable, reliable, and comprehensive target recognition performance.

In addition, Fig. 9 illustrates the statistical histogram distribution of the one-shot and five-shot recognition accuracy of Mada-SGD across various experimental configurations involving different depression angles. In the one-shot testing task, the recognition accuracy rates exhibit a relatively scattered and evenly distributed pattern among the four experimental configurations, ranging from a minimum accuracy rate of 31.11% to a maximum of 100%. for the five-shot testing task, the recognition accuracy demonstrates a distribution range between 51.11% and 100%, with a higher concentration of tasks falling within the higher accuracy interval, thereby enhancing recognition results in terms of concentration and efficiency. This observation suggests that as more samples are presented to the model, its recognition performance gradually improves while experiencing less fluctuation in accuracy, leading to enhanced stability in recognition performance. These findings are further supported by Table IV's experimental results, which serve as additional evidence for affirming that our proposed Mada-SGD is an enduring and dependable SAR-FSL target recognition model capable of effectively handling variations caused by depression angle under extended operational conditions.

2) Comparison With Existing SAR-FSL Methods: To comprehensively evaluate the proposed Mada-SGD method for few-shot SAR target recognition, this section conducts a series of comparative experiments and discussions with the latest available SAR-FSL target recognition models. The compared models mainly include two types of SAR-FSL methods: optimization-based and metric-based, including MAML [21], Reptile [23], Meta-SGD [24], MSAR [41], AG-MsPN [37], HIN [38], and MGA-Net [39]. Table VI lists the test results of Mada-SGD and the other SAR-FSL approaches. As shown in the experimental results, when testing tasks involve varying depression angles (ranging from 15° to 45°), each model achieves its highest at 17° but gradually decreases at 30° and 45°. However, our proposed Mada-SGD outperforms other state-of-the-art SAR-FSL methods in both the one-shot and five-shot testing tasks, demonstrating its effectiveness and reliability. Moreover, analyzing the experimental results presented in Table VI reveals that optimization-based methods exhibit stronger robustness to the depression variation than metric-based methods while achieving slightly higher target recognition accuracy. This can be attributed to the fact that metric-based SAR-FSL methods struggle to consistently represent the SAR target in the high-dimensional feature space as depression angle change significantly. On the other hand, optimization-based methods leverage good initialization prior knowledge which is then updated through inner loop updates for fast adaptation to new targets, effectively learning discriminative features of SAR targets thereby enhancing target recognition performance.

In addition, Fig. 10 illustrates the recognition accuracy and loss error curves of Mada-SGD in the 17°D/17°D testing task for both the one-shot and five-shot scenarios. It can be seen that as episode training progresses, the model's test accuracy gradually improves while the loss error steadily decreases. Specifically, among the five optimization-based SAR-FSL methods (MAML, Reptile, Meta-SGD, MSAR, and Mada-SGD), MAML and Reptile solely focus on optimizing a good

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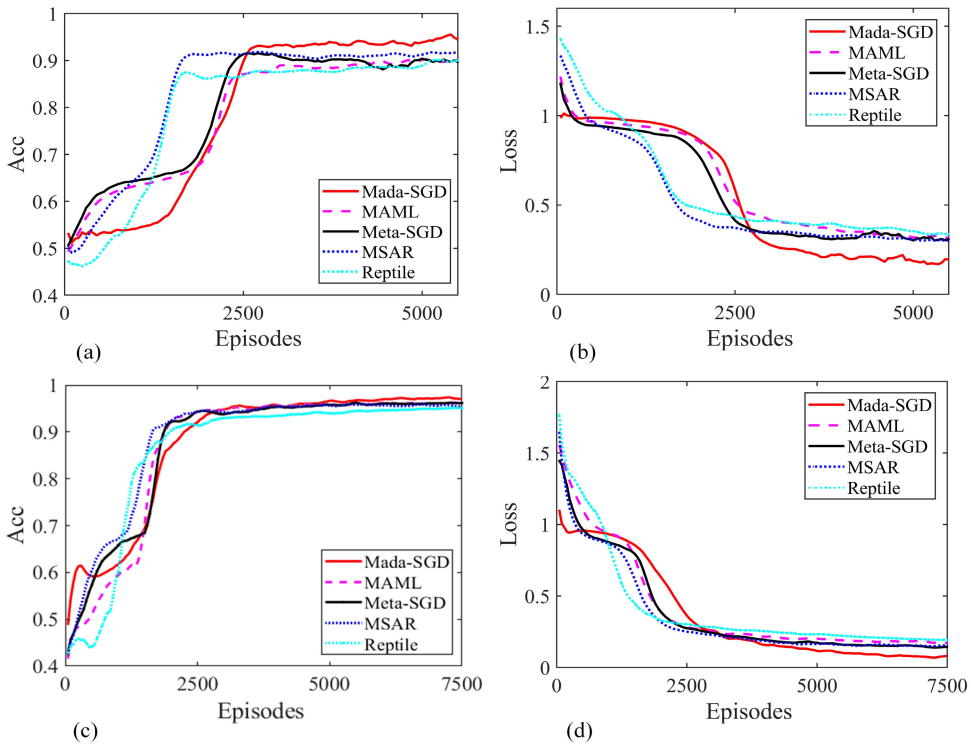


Fig. 10. Test accuracy and loss curves versus episodes. (a) and (b) 17°D/17°D (one-shot). (c) and (d) 17°D/17°D (five-shot).

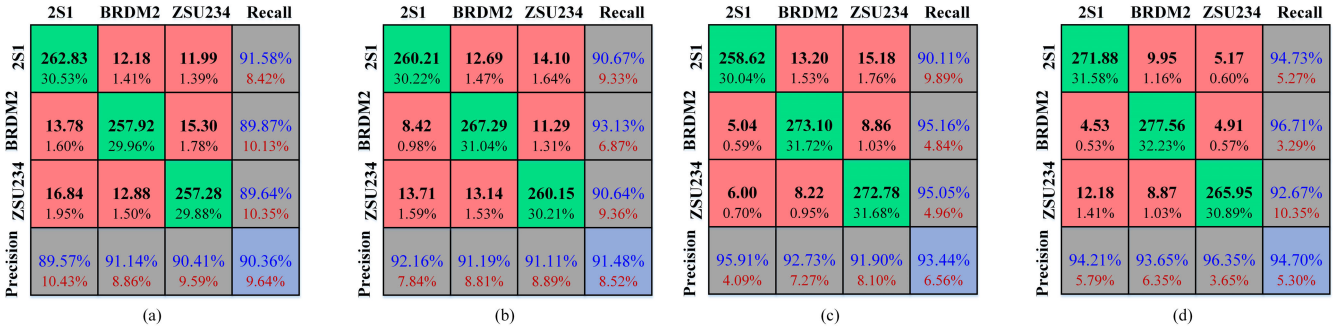


Fig. 11. Average confusion matrix. (a) Layerwise (one-shot). (b) Elementwise (one-shot). (c) Layerwise (five-shot). (d) Elementwise (five-shot).

initialization parameter. On the other hand, Meta-SGD and MSAR not only learn the initialization parameters but also optimize the learning rate in their update strategy. In contrast to these approaches, our proposed method Mada-SGD introduces a novel inner loop parameter updating mechanism that effectively adapts to learn initialization parameters, weight factors, update factors, and update directions to enhance feature representation capability and enable fast adaptation for few-shot targets. This multiparameter learning and optimization approach necessitates more episodes for training, updating, and adaptation. However, through effective parameter learning and updating processes, Mada-SGD demonstrates superior SAR target learning and adaptation capabilities compared with other models by achieving lower error losses along with higher SAR target recognition accuracy. Furthermore, when comparing results from both the one-shot and five-shot testing tasks alongside previous state-of-the-art approaches' performance improvements on four testing tasks (one-shot: 1.14%, 3.04%, 1.38%, and 2.70%; five-shot: 0.58%, 0.85%, 0.49%, and 2.12%), it becomes apparent that Mada-SGD exhibits more pronounced enhancements in performance for the one-shot testing task when compared with the results obtained from five-shot testing. This clearly highlights Mada-SGD's remarkable few-shot target learning abilities and its effectiveness in addressing few-shot SAR target recognition challenges.

C. Model Analysis

The purpose of this section is to investigate the impact of various components in Mada-SGD on the SAR-FSL target recognition performance. To explore its performance limits, a series of model analysis experiments are conducted. The details are presented as follows.

1) Multilevel Inner Loop Update Analysis: This article proposes a multilevel inner loop update strategy to address the varying task requirements in SAR-FSL and flexibly select suitable update strategies in practical applications to

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TABLE VI
COMPARATIVE TEST ACCURACY RESULTS OF MADA-SGD AND OTHER EXISTED SAR-FSL APPROACHES

Task Type	Models	Type	Acc (%)			
			17°D/15°D	17°D/17°D	17°D/30°D	17°D/45°D
3-way 1-shot	MAML [21]	Optimization-based	89.27%	90.90%	86.77%	68.23%
	Reptile [23]	Optimization-based	88.91%	90.49%	86.42%	68.38%
	Meta-SGD [24]	Optimization-based	89.56%	91.45%	87.06%	69.89%
	MSAR [41]	Optimization-based	90.02%	91.78%	87.62%	70.67%
	AG-MsPN [37]	Metric-based	90.84%	92.04%	87.80%	65.36%
	HIN [38]	Metric-based	89.73%	91.76%	87.82%	66.11%
	MGA-Net [39]	Metric-based	91.38%	91.87%	88.16%	68.98%
	Mada-SGD (Ours)	Optimization-based	**92.52%**	**95.08%**	**89.54%**	**72.59%**
3-way 5- shot	MAML [21]	Optimization-based	95.16%	96.19%	91.72%	74.70%
	Reptile [23]	Optimization-based	94.80%	95.92%	91.49%	74.23%
	Meta-SGD [24]	Optimization-based	95.08%	96.13%	91.90%	76.76%
	MSAR [41]	Optimization-based	95.61%	96.42%	92.25%	77.27%
	AG-MsPN [37]	Metric-based	95.33%	95.72%	91.93%	73.04%
	HIN [38]	Metric-based	94.99%	95.54%	91.59%	72.53%
	MGA-Net [39]	Metric-based	95.40%	96.17%	92.21%	75.47%
	Mada-SGD (Ours)	Optimization-based	**96.19%**	**97.27%**	**92.74%**	**79.39%**

TABLE VII
TEST RESULTS OF MULTILEVEL INNER LOOP UPDATE

Model	Base Learner	Experiment Type		Accuracy (%)		
		Dataset	Task	Model-wise	Layer-wise	Element-wise
Mada-SGD	ConvNet4	17°D/30°D	1-shot	89.54	90.36	91.48
			5-shot	92.74	93.44	94.70

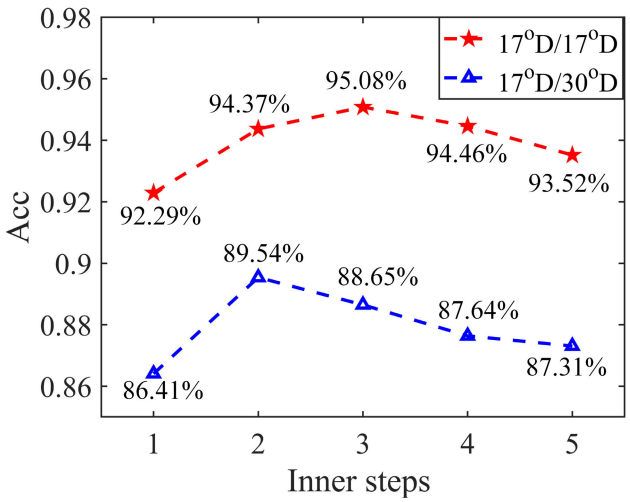


Fig. 12. Influence of inner steps on the test accuracy with 17°D/17°D (one-shot) and 17°D/30°D (one-shot) experiment types.

TABLE VIII
ABLATION STUDY OF WEIGHT FACTOR AND UPDATE FACTOR

Experiment Type	Weight Factor	Update Factor	Acc
17°D/30°D 1-shot	×	×	86.27%
	✓	×	87.84%
	×	✓	88.79%
	✓	✓	**89.54%**

× represents that the learnable parameters remain fixed and unchanged throughout the updating process, while ✓ indicates that the learnable parameters are adaptive and undergo changes during the updating process.

avoid issues such as high model complexity and overfitting. This experiment primarily investigates the impact of multilevel inner loop update strategies on the target recognition performance of Mada-SGD, including three scales of update rules: modelwise, layerwise, and elementwise. Table VII presents the experimental results of multilevel inner loop update strategies on the 17°D/30°D testing tasks. It can be observed that in the one-shot testing tasks, the recognition accuracy for each of the three update rules is: modelwise (89.54%), layerwise (90.36%), and elementwise (91.48%). In addition, for the five-shot testing tasks, there is further improvement in Mada-SGD's recognition performance with a maximum SAR-FSL target recognition accuracy of 94.70%. Fig. 11 exhibits the average confusion matrix of layerwise and elementwise inner loop update rules. By combining these experimental results with those presented in Table V, it can be concluded that Mada-SGD has achieved favorable SAR-FSL target recognition results under all the three hyperparameter update rules while demonstrating high recognition accuracy across all types of targets considered here. Through analysis of these experimental results, it becomes evident that improvements in recognition performance mainly stem from enhanced recall rates for two specific types of targets: BRDM2 and ZSU234. As learnable parameters increase, Mada-SGD

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TABLE IX
TEST RESULTS OF DIFFERENT KERNEL SIZES IN BASE LEARNER

Model	Base Learner	Experiment Type		Kernel Size	Parameters (K)	Accuracy (%)
		Dataset	Task			
Mada-SGD	ConvNet4	17°D/30°D	1-shot	3×3	116.16	89.54
				5×5	315.84	**89.93**
				7×7	615.36	87.92
				9×9	1014.72	88.41

allocates more attention to BRDM2 and ZSU234 targets resulting in a rapid decline in misclassification rate; thus validating the effectiveness of our proposed multilevel inner loop update strategy outlined within this article. In practical applications, selecting appropriate parameter update strategies based on specific experimental task requirements is crucial for achieving a balance between performance and model complexity.

2) Hyperparameters Analysis: The impact of weight factors β and update factors η in Mada-SGD on the recognition performance of SAR-FSL is primarily analyzed and discussed in this section. Table VIII presents the results of ablation experiments conducted on these hyperparameters. From the experimental results, it can be observed that when the weight factors are kept fixed without being involved in parameter iteration updates, the recognition accuracy reaches 86.27%. When only the weight factors are activated while keeping the update factors fixed, the recognition result improves to 87.84%. Similarly, when only the update factors are activated while keeping the weight factors fixed, a final recognition result of 88.79% is achieved. However, by simultaneously activating both weight and update factors adaptively in Mada-SGD, we achieve a highest SAR target recognition performance of 89.54%. Comparing these results with separate adaptive activation experiments for each hyperparameter reveals that although both have positive effects on recognizing SAR targets with few samples, it is evident that update factor has a greater impact on improving target recognition performance. This may be attributed to its direct influence on determining both the direction and step size for each parameter update during updating steps, whereas weight factor considers multiple previous and subsequent updated step parameters comprehensively resulting in a slightly delayed effect on updated step parameters. In summary, these ablation experimental results fully validate effectiveness and reliability of the proposed Mada-SGD and highlight the positive contribution played by introduction of weight factor β and update factor η for recognizing SAR targets with few samples.

3) Effectiveness of Update Steps and Hyperparameters: Mada-SGD introduces a meta-adaptive hyperparameter update strategy to enhance the feature representation ability for few-shot SAR targets by learning the correlation information between multiple inner loop update steps of parameters. Specifically, this is achieved by incorporating learnable weight factors and update factors that dynamically scale and update the parameters during each inner loop iteration, enabling the retention of correlation information across multiple steps. In this section, we primarily analyze and discuss the number of update steps in the inner loop multistep memory mechanism and the distribution of weights among these update steps. Fig. 12 depicts the impact of different inner loop update steps on the recognition accuracy of SAR-FSL using Mada-SGD in 17°D/17°D and 17°D/30°D experiment types. It can be observed that in the 17°D/17°D (one-shot) experiment type, Mada-SGD achieves the highest recognition accuracy of 95.08% at inner steps of three. On the other hand, in the 17°D/30°D (one-shot) experiment type, Mada-SGD achieves a peak recognition accuracy of 89.54% at inner steps of two. This indicates that there is no fixed optimal number of inner steps for Mada-SGD across different experiment types; instead, it should be trained and learned to determine the best number of inner steps based on specific experimental requirements.

Furthermore, Fig. 13 illustrates how Mada-SGD updates its hyperparameters, including weight factors, update factors, and weight distributions under optimal performance conditions for each mentioned experiment type. As training progresses across episodes, weight factors and update factors gradually deviate from their initial values to adapt to different learning tasks [Fig. 13(a), (b), (d), and (e)]. Fig. 13(c) and (f), respectively, demonstrates weight distribution between inner steps under optimal performance conditions for both the experiments mentioned above; slight differences can be observed in weight distribution among different experimental tasks. However, Mada-SGD consistently assigns learning weights to each initialization parameter in every inner step, ensuring that the contribution of each updated parameter is reflected in the final recognition result. By considering multiple updated parameters instead of solely relying on a single updated step parameter alone, SAR-FSL's recognition performance is further improved as it helps learn correlation information among them. This fully validates the advancement and effectiveness brought by our proposed Mada-SGD.

4) Base Learner Convolution Kernel Size Analysis: This section mainly investigates the influence of convolution kernel size on the performance of Mada-SGD in few-shot target recognition. In this experiment, we configure four groups of convolutional kernels 3×3, 5×5, 7×7, and 9×9, for the entire base learner to acquire varying scales of receptive fields. And the input SAR image size remains fixed at 64×64. The experimental results are shown in Table IX. It can be observed that the recognition accuracy for each convolution kernel sizes is 89.54%, 89.93%, 87.92%, and 88.41% (kernel size from smaller to larger). Notably, Mada-SGD achieves

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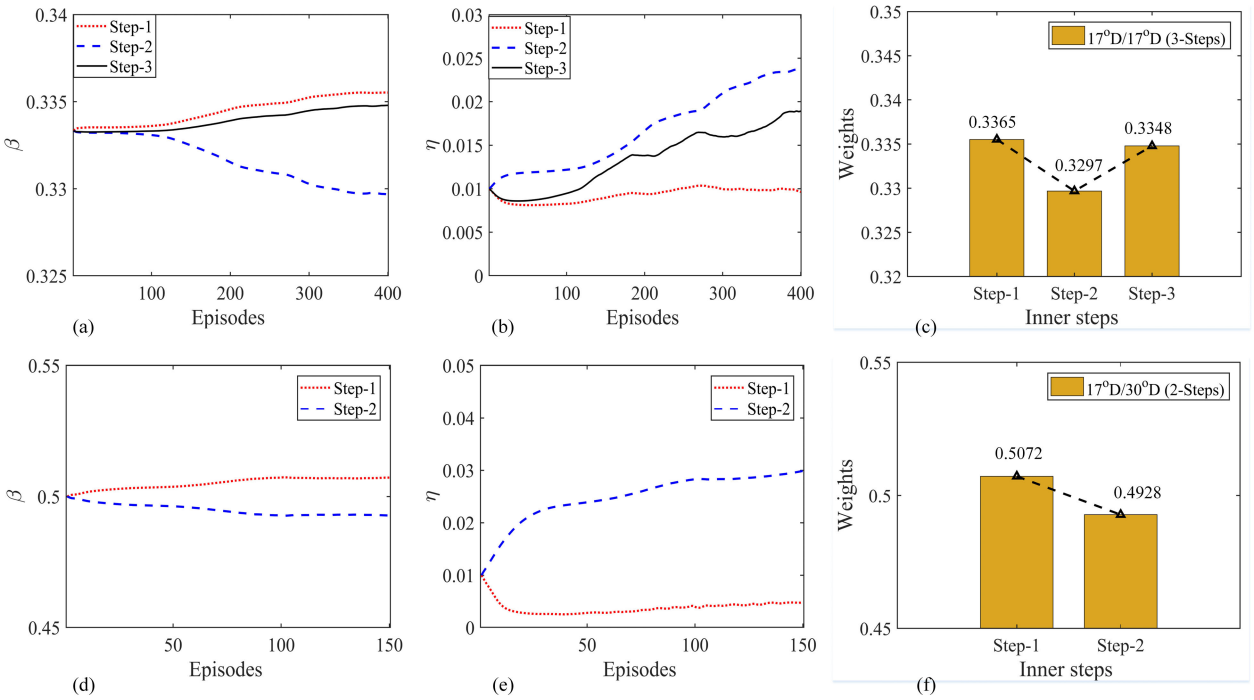


Fig. 13. Update of inner loop hyperparameters in Mada-SGD. (a) Weight factors (17°D/17°D-1-shot). (b) Update factors (17°D/17°D-1-shot). (c) Weight distribution under three inner loop updating steps (17°D/17°D-1-shot). (d) Weight factors (17°D/30°D-1-shot). (e) Update factors (17°D/30°D-1-shot). (f) Weight distribution under two inner loop updating steps (17°D/30°D-1-shot).

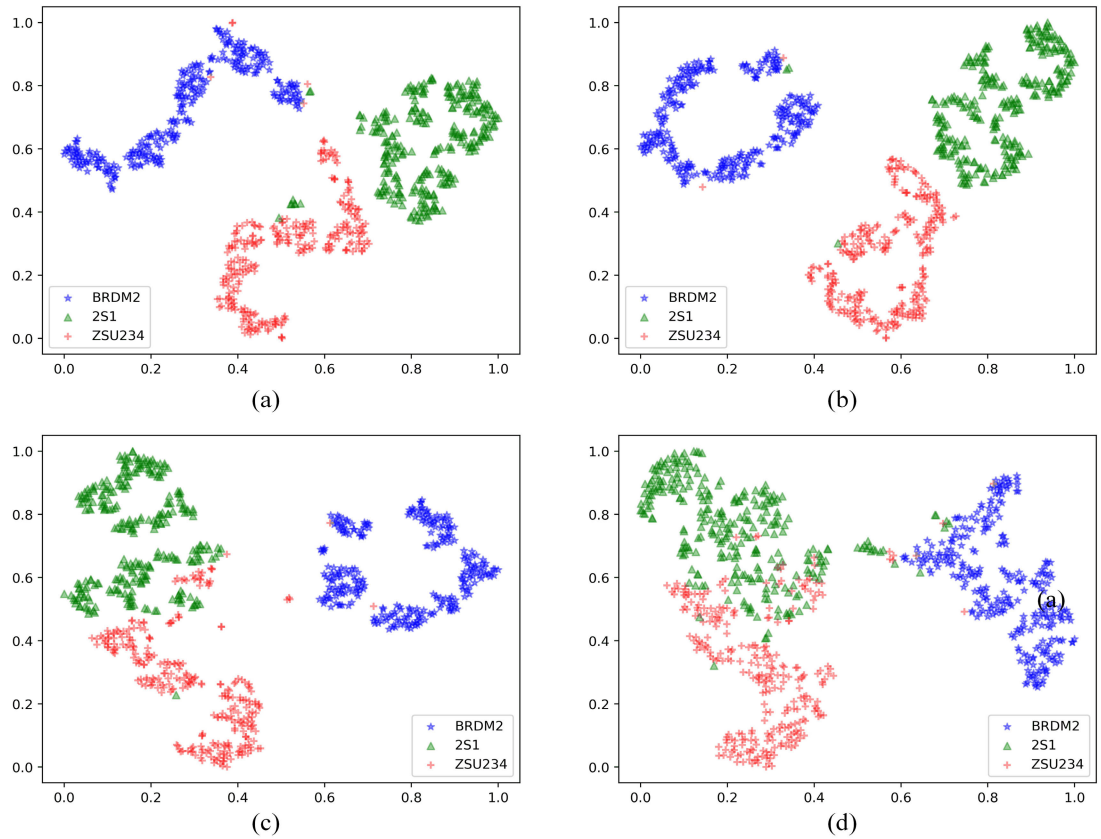


Fig. 14. Visualization of 2S1, BRDM2, and ZSU234 targets under different kernel size by t-SNE dimensionality reduction. (a) Kernel size = 3. (b) Kernel size = 5. (c) Kernel size = 7. (d) Kernel size = 9.

its highest SAR-FSL recognition accuracy when using a convolution kernel size of 5. Conversely, it obtained its lowest recognition rate when using a size of 7. This phenomenon can be attributed to smaller kernel sizes (3 or 5) effectively capturing local texture features of the targets while larger ones (7 or 9) prioritize macroscopic

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features at a larger scale. For small-size SAR target images (64×64 used for training and testing purposes), convolutional kernel sized at 3 or 5 proves more advantageous in extracting discriminative features and improving representation for few-shot SAR targets. Furthermore, Fig. 14 depicts the t-distributed stochastic neighbor embedding (t-SNE) visualization results of the meta-testing set targets, namely, 2S1, BRDM2, and ZSU234, across different convolutional kernel size settings. Evidently, the test targets exhibit a more compact and distinguishable feature space distribution when using kernel sizes of 3 and 5. Conversely, with kernel sizes of 7 and 9, the targets appear sparsely distributed with a pronounced aliasing phenomenon leading to decreased recognition performance stability. Therefore, Mada-SGD adopts a baseline model with convolutional kernel size of 3 to achieve high intra-class similarity and interclass difference while minimizing model parameters for improving stability and reliability in SAR-FSL target recognition.

V. Conclusion

This article proposes a meta-adaptive hyperparameter update method, named Mada-SGD, for optimizing inner loop fast adaptation in few-shot target recognition under severely limited samples in SAR-ATR. Mada-SGD introduces a meta-adaptive hyperparameter update strategy to automatically learn initialization parameters, weight factors, update factors, and update directions in the meta-learner. It aims to quickly generalize prior knowledge from the meta-training task to effectively represent few-shot SAR target features and improve the efficiency of inner loop update and optimization. In additional, we use weight factors to scale the parameters of multiple update steps similar to a memory mechanism to leverage correlation information between inner loop updates. This further enhances Mada-SGD's feature representation ability for SAR targets. Moreover, Mada-SGD provides an effective and feasible SAR-FSL method that addresses the current lack of research on optimization-based meta-learning methods in the SAR-FSL field. It contributes toward improving the theoretical system of SAR-ATR. Future potential research work mainly focuses on adaptive iteration research of inner loop multistep update optimization and feasibility analysis of model-based few-shot SAR target recognition methods.

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